



Power Systems Business Group Asset Management Department

Load Assessment Procedure

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Document No. : PS-COP-0215

Revision No. : 07

Endorsed by:

A blue ink signature, appearing to be "D. A.", written over a horizontal line.

Senior Network Planning Manager

Date: 15 Sept 2012

Approved by:

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Deputy Director – Asset Management

Date: 17 Sep 2012

Load Assessment Procedure

Table of Changes

Version	Date	Changes to previous version
Rev. 05	September 2006	General revision. Parameters remain unchanged.
Rev. 06	September 2009	Para 3.3.5 added with ADMD for office (including air conditioning load) revised to 0.26 kVA/m ² . Para 3.6 on MDDE revised.
Rev. 07	September 2012	Parameter for flatted factory deleted. Parameter for Data Centre newly added. Example in Appendix 6 corrected.

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1. Introduction

The purpose of Code of Practice (COP) 215 is to standardize the load assessment procedure and the After Diversity Maximum Demand (ADMD) figures in two Regions of the Company so as to facilitate planning engineer in designing adequate 11kV/LV supply capacity for various property developments. The first edition of COP215 was issued in 1983. Subsequently the document was revised in 1993, 2000, 2003, 2006 and 2009 in order to cope with the changes during 1980s to 2000s. Provision on new power requirements such as electric water heating, electric kitchen, etc. were incorporated since revision 3. In order to reflect the latest consumption pattern and to continuously improve the process, this document is reviewed periodically.

2. Objectives & Scope

2.1 Objectives

To ensure that the loading demands of property developments are estimated as accurately as possible for provision of adequate supply capacity in order to avoid: -

- Under-estimation - resulting in the sufferings of normal business operations and the inconvenience of residents in the developments
- Over-estimation - resulting in the installation of under-utilised capacity and investment.

In order to arrive at reliable results, all estimations should be reviewed in conjunction with the load research data and sufficient margin should be reserved in accordance with the latest market conditions to cater for forthcoming load growth.

2.2 Scope

This document serves as guidelines for the planning engineers who participate in load estimation for property developments or related work. It details various steps during the estimation of electricity demand for any project, large or small, upon which the Company makes its investment decisions on what supply capacity should be installed.

Broadly speaking, the load estimation has been classified under the following major headings: -

- Residential
- Commercial
- Industrial
- Municipal

For mixed development, such as Residential-Commercial, Commercial-Industrial, etc, Mixed Development Diversity Factors are also included for load estimation.

3. Guidelines for Load Assessment

3.1 General

3.1.1 Major Classifications

For load assessment purpose, property developments are broadly classified as 'Residential', 'Commercial', 'Industrial', 'Municipal', or a mix of them.

3.1.2 Usable Floor Area

Floor Area is the basic parameter to calculate the loads of developments. Among the existing floor area definitions, Usable Floor Area / Space (UFA), which is defined by Building (Planning) Regulations is adopted for our load assessment purpose.

Definition of Usable Floor Area / Space (UFA):

'Usable Floor Space' means the aggregate of the areas of the floor or floors in a storey or building excluding any staircases, public circulation space, lift landings, lavatories, water-closets, kitchens, and any space occupied by machinery for any lift, air-conditioning system or similar service provided for the building.

Since Building Department requests developers to submit UFA information of property developments to them, it will not be difficult for a planning engineer to request the said information from an applicant if it is not provided in an application for electricity supply.

3.1.3 Assessed Load

All load estimates are based on 'Assessed Load' only, which will be used for both transformer room reservation and transformer allocation. The 'Anticipated Load' figures are not applicable in this procedure.

3.1.4 Development Information

In preparing preliminary supply scheme, planning engineer should request the applicant to submit detailed development information such as floor plan with UFA, floor area usage and loading information including public services and any special loads. To facilitate planning engineer for subsequent data inputting to the LAST (Load Assessment Programme), the applicant is requested to fill in the standard load estimation sheet shown in Appendix 8. If central air conditioning and/or central electric water heating systems are adopted for a development, applicant should also submit additional loads details as well shown in Appendix 9. A flow chart for estimation of assessed load is also shown in Appendix 10.

3.1.5 Load Assessment Programme (LAST)

Having collected the required development information, the planning engineer should input the corresponding information into a computer programme called Load Assessment Programme (LAST) to estimate the total 'Assessed Load'. From the results, the required number of distribution transformers and number of transformer bays can be determined. The LAST database should be properly maintained in order to cater for future reference and analysis.

3.1.6 Public Services

All ADMD figures do not include public services load and it is necessary to assess it independently. Reference figures on per lift basis are provided for residential, commercial and industrial developments and the figures have already included public lighting, lift, water pump, fire services and lobby air conditioning. Owing to the great variety of the rating of the individual installation, applicant's load estimations for public services shall be solicited for crosschecking with the reference figures.

3.1.7 Multi-Transformer Case

For development with multi-transformers serving for different zones, the final total assessed load should be multiple by 1.1.

3.1.8 LV Outfeed Loads

For any developments, the inputting to the LAST should not include any LV outfeed loads, since this will cause confusion in future analysis of the LAST data.

3.2 Residential Development (Please refer to Appendix 1)

3.2.1 Housing Type Classification

As depicted in the loading data, the loading demand patterns for public rental and private housing are getting similar. Hence, all classifications are merged into a single category, i.e. 'All Housing Development', which includes all types of permanent housing, like Public Rental Housing, Home Ownership Scheme (HOS), Private Sector Participation Scheme (PSPS), Village Housing and Private Housing.

3.2.2 Residential ADMD Figures

Residential ADMD figures are presented in per flat basis. The exact figure selected depends on both the UFA of the flats and the number of flats served by the same supply. Three classes of UFA are chosen, namely, (1) less than 50 m², (2) between 50-80 m² and (3) above 80 m². Taking into consideration of diversity effect, for a development of less than 50 flats, a range of figures is used for the same UFA class depending on the number of flats served. For a development of more than 50 flats, a single value is used on the same UFA class. All residential ADMD figures have already included all typical lighting, general power and instantaneous electric water heating loads. For fixed electric cooking installation, individual assessment is still required.

3.2.3 Public Services and Clubhouse

Public services and clubhouse loads should be assessed independently. For public services, a reference figure of 55 kVA/lift is adopted for easy reference and the figures have already included public lighting, lift, water pump, fire services and lobby air conditioning. For development without lift service, a minimum provision of 10 kVA of public services load should be provided. Owing to the great variety of

the power rating of the individual installation, applicant's load estimations for public services and clubhouse shall be referenced.

3.2.4 Developments with Mixed UFA Flats

For developments with mixed UFA flats, planning engineer should first identify the total number of mixed flats, and then read the corresponding ADMD figures. For instance, if the development contains 8 flats with $\text{UFA} < 50\text{m}^2$ and 10 flats with $\text{UFA} > 80\text{m}^2$, the figures for 18 number of flats should be selected, i.e. 7.6 kVA/flat for $\text{UFA} < 50\text{m}^2$ and 14.5 kVA/flat for $\text{UFA} > 80\text{m}^2$ respectively.

3.2.5 Special Flats

For multi-floor flats, planning engineer should decide the equivalent number flats for the assessment, e.g. a duplex flat occupying 70 m^2 per floor, it should be counted as two flats of UFA between $50\text{-}80\text{ m}^2$. Besides, for $\text{UFA} > 150\text{ m}^2$, individual assessment is required.

3.3 Commercial Development (Please refer to Appendix 2)

3.3.1 Commercial ADMD Figures

All the ADMD figures are represented in per m^2 basis. All ADMD figures have not included central air conditioning load and central electric water heating loads. Hence, individual assessment is required. The proposed figure of 40 kVA/lift is just a reference for general public services and it should be assessed individually whenever possible.

3.3.2 Commercial Complex

'Commercial Complex' – refers to large-scale shop floor areas that cannot be determined as Catering or Non-Catering at the time when an applicant submits application for supply with a development plan. It is assumed that in general about 30% floor area of the commercial complex will be used for catering purposes.

3.3.3 General Shop

'General Shop - Catering' – refers to restaurants, cafes, fast food shops, and Chinese restaurants.

'General Shop - Non-Catering' – refers to any shops other than catering shops. The ADMD figures for both catering and non-catering shop have already included non-central air conditioning load and 0.1 kVA/m^2 has to be deducted in case there is central air conditioning system, which will be assessed separately.

3.3.4 Hotel

In order to estimate the load more accurately, hotel area is divided into 3 types for load assessment, namely restaurant, shopping mall and accommodation area. Different figures will be adopted for these areas.

3.3.5 Office

As the ADMD figure has not included central air conditioning load, individual assessment for central air conditioning is required. In case central air conditioning load is not available, $0.1(\text{kVA/m}^2)$ should be added for the development.

3.4 Industrial Development (Please refer to Appendix 3)

3.4.1 Industrial ADMD Figures

All industrial ADMD figures are for reference only. As the industrial applications vary greatly, the planning engineer has to secure load details from the load information initially enclosed in an application letter or by using the load estimation sheet as shown in Appendix 7 whichever is appropriate before carrying out load assessment. He may need to consult and discuss with the applicant in case of significant variation between the two parties' load estimates.

For application, when no central but window type air conditioners are anticipated, 0.1 kVA/m^2 could be added on top of the ADMD figures. A figure of 50 kVA/lift is a reference for general public services and it should be assessed separately when information available.

As there is rarely flatted factory buildings completed recently or under construction, nor new flatted factory projects likely in future, its load parameter is deleted.

3.5 Data Centre Development (Please refer to Appendix 4)

Numerous data centres have been completed recently and new ones are under construction. The data centres usually require large power supply capacities. To facilitate load assessment for data centres, new parameters have been added as indicated in Appendix 4.

3.6 Municipal Development (Please refer to Appendix 5)

3.6.1 Municipal ADMD Figures

ADMD figures for 'General Municipal Development' and 'School & College' have already included non-central air conditioning load and 0.1 kVA/m^2 has to be deducted in case there is central air conditioning system in the development. However, all ADMD figures have not included public services, central air conditioning and central electric water heating loads. Hence, individual assessment is required for such loads.

3.6.2 General Municipal Development Classification

The classification refers to Youth Centre, Home for Elderly, Community Centre, Indoor Game Hall, Urban Market Complex, Fire Station, Police Station, Ambulance Depot, etc.

3.6.3 School & College

The classification refers to Kindergarten, Primary, Secondary, and Post-Secondary Schools with air conditioning and IT provision.

3.7 Mixed Development Diversity Factors - MDFF (Please refer to Appendix 6)

Mixed Development Diversity Factors (MDFF) are provided in the LAST but planning engineer can override these MDFF figures by using his own judgement as necessary. From load research data, it is revealed that the peaks for both commercial and industrial are nearly coincident. Hence, MDFF for Commercial-Industrial is 1.0 for all mixed. As for Residential-Commercial, the latest MDFF figures are given in the Appendix 6. Recent experiences indicate that residential towers and commercial shopping areas in large-scale developments are planned to be supplied by dedicated transformers. Planning engineers should be aware of this in applying the MDFF.

As data center has flat load profile and municipal development has irregular load profile, both have unity MDFF.

3.8 Central Air Conditioning Load (Please refer to Appendix 7)

As central air conditioning load is not included in the ADMD figures, it should be separately assessed. A table in Appendix 7 indicates the electrical power conversion figures of some typical central chilling systems. This table can provide some means to crosscheck the central air conditioning load claimed by an applicant. From the information sheet for central air conditioning load estimation submitted by the applicant, the calculated electrical power conversion figures (as obtained by dividing the total electrical air conditioning load by the number of tonne refrigeration required) can be compared with those in Appendix 7. If the figures are close, the estimated air conditioning load provided by the applicant can be inputted into the LAST. Otherwise, the planning engineer has to discuss with the applicant in order to validate their estimated air conditioning load.

3.9 Other Special Loads

Similar to central air conditioning, the power requirement of other special loads requires separate assessment, e.g. central electric water heating system. Therefore, planning engineer should solicit relevant information and discuss with the applicant on their load estimation.

4. Load Assessment Procedure

The load assessment procedure should be performed by the planning engineer with the following details of activities.

- 4.1 Architect/consultant/developer submits application for supply letter together with building floor plan, relevant loading information and standard load estimation sheets.
- 4.2 For typical development categories, planning engineer can proceed with the assessed load estimation by using the LAST. If a development involves central air conditioning, central electric water heating systems and/or any other special loads,

the applicant must submit individual load details so that the planning engineer could have adequate information to assess the total load. The accepted load estimate for the special load should then be inputted into the LAST as a separate load item and thus it forms part of the total assessed load.

- 4.3 To compare the assessed load with applicant's declared load in 4.1 and 4.2 and check the variance between the two parties' load estimates. The load estimates are considered acceptable if
- 1) The load estimates from the applicant is greater than those from the planning engineer, or
 - 2) The variance does not affect the overall distribution transformer room requirements.
- 4.4 If the load estimates from the applicant is smaller than those from the planning engineer and the variance will cause the installation an additional transformer, consultation with the applicant is required to identify the causes of the variance. However, the assessed load compiled from the LAST should be adopted unless reasonable justifications with design/technical details are provided by the developer/consultant. Planning engineer should revise the load estimation, if necessary, and update the LAST records accordingly for any changes.
- 4.5 After finalising the total assessed load, planning engineer can determine the number of transformers and transformer bays required for that development.
- 4.6 All the records in the LAST should be properly kept in the server so that relevant parties in PSBG and MACS could assess the database for future reference and analysis.

5. Applications of ADMD Figures

ADMD figures are load densities derived by dividing the aggregate load readings by the corresponding usable floor areas (non-residential development) or the number of flats (residential development) fed by them. Since load densities of developments of the same category vary considerably in magnitude, the ADMD figures are so chosen that they can cover 85% of the developments depend on their categories. Together with the spare

capacity of the allocated transformers, this implies that it should be safe enough for most of the planning cases.

However, in very few cases, the load estimates may come out to be marginal for transformer room reservation or transformer allocation, then the planning engineer has to exercise his own discretion whether to adjust the load estimates for offsetting the under estimation risk or not.

Besides, the figures provided in this document are designed primarily for planning the supply capacity, e.g. transformer allocation and transformer room reservation. The figures may not be suitable for designing LV installations after the customer mains switch, e.g. raising mains and lateral mains.

6. Review and Future Update

As to reflect the latest market conditions, a review will be conducted before September 2015. In case there are significant changes in the market conditions during the period, review on particular parameter could be carried out upon request.

Appendix 1 – ADMD Figures for Residential Developments

<u>Classification</u>	<u>ADMD Figures (kVA/Flat)</u>			
	<u>1-10 Flats</u>	<u>11-20 Flats</u>	<u>21-50 Flats</u>	<u>≥ 50 Flats</u>
All Housing Development¹				
UFA: < 50 m ²	31.2 – 9.8	9.8 – 6.9	6.6 – 4.0	3.9
UFA: 50 - 80 m ²	39.6 – 16.1	16.1 – 11.4	10.9 – 6.6	6.5
UFA: > 80 m ²	52.8 – 18.6	18.6 – 13.4	12.8 – 8.0	7.8
Car Park with Ventilation	0.03 / m ²			
Public Services²	55 kVA/lift			

Footnotes:

1. 'All housing development' refers to all types of permanent housing, like Public Rental Housing, Home Ownership Scheme, Private Sector Participation Scheme, Village Housing and Private Housing.

ADMD Figures have already included typical lighting, general power and instantaneous electric water heating loads. For fixed electric cooking installation, public services and clubhouse loads, individual assessment is still required.

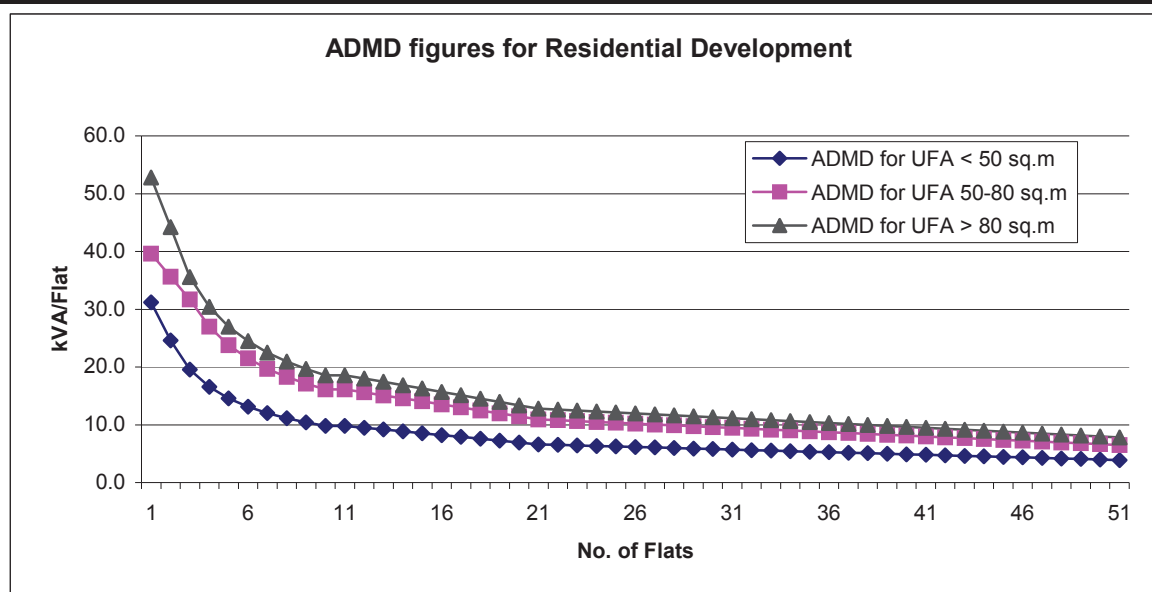
For development with central air conditioning or central electric water heating system, planning engineer should decide whether the figures are appropriate for the assessment.

For multi-floor flat, planning engineer should decide the number of equivalent flats for assessment. For UFA > 150 m², individual assessment is required.

2. Figures for 'Public Services' includes public lighting, lift, water pump, fire services and lobby air conditioning. For development without lift service, it is subject to the E&M submission and a minimum provision of 10 kVA of public services load should be provided.

Appendix 1 – ADMD Figures for Residential Developments (Detailed Table)

No. of Flats	ADMD for UFA < 50m ²	ADMD for UFA 50-80m ²	ADMD for UFA > 80m ²
	kVA/flat	kVA/flat	kVA/flat
1	31.2	39.6	52.8
2	24.6	35.7	44.2
3	19.5	31.7	35.5
4	16.6	27.0	30.4
5	14.6	23.8	27.0
6	13.1	21.5	24.5
7	12.0	19.7	22.5
8	11.2	18.3	21.0
9	10.4	17.1	19.7
10	9.8	16.1	18.6
11	9.8	16.1	18.6
12	9.5	15.6	18.0
13	9.2	15.1	17.4
14	8.9	14.6	16.8
15	8.5	14.0	16.3
16	8.2	13.5	15.7
17	7.9	13.0	15.1
18	7.6	12.5	14.5
19	7.3	12.0	14.0
20	6.9	11.4	13.4
21	6.6	10.9	12.8
22	6.5	10.8	12.6
23	6.4	10.6	12.5
24	6.3	10.5	12.3
25	6.2	10.3	12.1
26	6.2	10.2	12.0
27	6.1	10.0	11.8
28	6.0	9.9	11.6
29	5.9	9.7	11.5
30	5.8	9.6	11.3
31	5.7	9.5	11.1
32	5.6	9.3	11.0
33	5.5	9.2	10.8
34	5.4	9.0	10.6
35	5.3	8.9	10.5
36	5.3	8.7	10.3
37	5.2	8.6	10.1
38	5.1	8.4	10.0
39	5.0	8.3	9.8
40	4.9	8.1	9.6
41	4.8	8.0	9.5
42	4.7	7.8	9.3
43	4.6	7.7	9.1
44	4.5	7.5	9.0
45	4.4	7.4	8.8
46	4.4	7.2	8.6
47	4.3	7.1	8.5
48	4.2	6.9	8.3
49	4.1	6.8	8.1
50	4.0	6.6	8.0
51	3.9	6.5	7.8



Appendix 2 – ADMD Figures for Commercial Developments

<u>Classification</u>	<u>ADMD Figures¹ (kVA/m²)</u>
Commercial Complex	0.30
General Shop - Catering ²	0.66
General Shop - Non-catering ²	0.40
Hotel - Restaurant	0.56
Hotel - Shopping Mall	0.30
Hotel - Accommodation	0.03
Office	0.16
Cinema	0.27/seat
Car Park with Ventilation	0.03
Public Services ³	40kVA/lift

Footnotes:

1. As ADMD figures have not included central air conditioning load and central electric water heating load, individual assessment is required for such loads.
2. ADMD figures for both 'General Shop – Catering' and 'General Shop – Non-catering' have already included non-central air conditioning load and 0.1 kVA/m² has to be deducted in case there is central air conditioning system covered in the development.
3. Public Services Load requires individual assessment and 40kVA/lift is a reference figure only.

Appendix 3 – ADMD Figures for Industrial Developments

<u>Classification</u>	<u>ADMD Figures¹ (kVA/m²)</u>
Godown	0.05
Cold-Storage Godown	0.05 + cooling plant load
Car Park with Ventilation	0.03
Public Services ²	50 kVA/lift

Footnotes:

1. All ADMD figures for Industrial Development are for reference purpose, planning engineer should make their judgement for individual cases.

When no central but window type air conditioners are anticipated, 0.1 kVA/m² could be added on top of those ADMD figures.

2. Public Services load requires individual assessment and 50kVA/lift is a reference figure only.

Appendix 4 – ADMD Figures for Data Centre Developments

<u>Classification</u>	<u>ADMD Figure¹ (kVA/rack)</u>	<u>ADMD Figure¹ (kVA/m²)</u>
Data Center	10.33	1.96

Footnote:

1. The ADMD figures have included public services and central air conditioning loads.

The ADMD figures for data centres vary greatly depending on the type of racks installed. The above figures are for reference purpose and planning engineer should make their judgement for individual cases.

Appendix 5 – ADMD Figures for Municipal Developments

<u>Classification</u>	<u>ADMD Figures¹ (kVA/m²)</u>
General Municipal Development ²	0.25
School & College ³	0.25
Hospital & Clinic	0.22
Car Park with Ventilation	0.03

Footnotes:

1. As ADMD figures have not included public services, central air conditioning load and central electric water heating load, individual assessment is required for such loads.
2. General Municipal Development includes Youth Centre, Home for Elderly, Community Centre, Indoor Game Hall, Urban Market Complex, Fire Station, Police Station, Ambulance Depot, etc. The ADMD figures have included non-central air conditioning load and 0.1 kVA/m² has to be deducted in case there is central air conditioning system in the development.
3. School & College includes Kindergarten, Primary, Secondary, and Post-Secondary Schools with air conditioning and IT provision and 0.1 kVA/m² has to be deducted in case there is central air conditioning system in the development.

Appendix 6 – Mixed Development Diversity Factor (MDDF)

Residential-Commercial Mixed

<u>% of Residential</u>	<u>% of Commercial</u>	<u>MDDF Figures</u>
10%	90%	1.06
20%	80%	1.12
25%	75%	1.15
30%	70%	1.19
40%	60%	1.25
50%	50%	1.22
60%	40%	1.19
70%	30%	1.16
75%	25%	1.13
80%	20%	1.11
90%	10%	1.05

Commercial-Industrial Mixed

MDDF: 1.00 for all mixes

Footnotes:

It has to be noted that the percentage mix is rounded off to 0, 25, 50 and 75% by the LAST automatically before applying the MDDF as shown in the above table. While keeping the MDDF in LAST unchanged, the following table for residential/commercial mix illustrates the principle of rounding off:

<u>Example</u>	<u>% Mix Range (Residential/Commercial)</u>		<u>MDDF</u>
	<u>Before rounding off</u>	<u>After rounding off</u>	
A	0/100-13/87	0/100	1.00
B	14/86-25/75	25/75	1.15
C	26/74-38/62	25/75	1.15
D	39/61-50/50	50/50	1.22

Appendix 7 – Electrical Power Conversion Figures for Typical Chilling Systems

Central Chilling System

Electrical Power Conversion Figures

(kW/Tonne Refrigeration)

Packaged air cooled	1.85
Indirect sea water cooled	1.65
Direct sea water cooled	1.45
Direct water cooled	1.45

Appendix 8 – Load Estimation Sheet for General Development

(to be completed by Electricity Supply Applicant)

Letter Reference		Submission Date		Supply Date	
Supply Address					
Developer		Contact Person / Title		Contact Telephone	Contact Fax
Contact Address					
Type of Development (Please tick in the box provided)					
☐ Residential: ☐ Public Rental ☐ HOS/PSPS ☐ Village Housing ☐ Private Housing ☐ Others _____ ☐ Commercial ☐ Industrial ☐ Municipal ☐ Others _____					
Tenant Supply		Usable Floor Area (m ²)	No. of Units	Average Unit Loading (kVA/m ²)	Sub-total Loading (kVA)
Floor Level Description	Designed Use				
Landlord Supply: Building Services / Central A/C / Central EWH / Car Park / Clubhouse / Others					
Remarks			Total Load (kVA)		
			Total Residential		%
			Total Commercial		%
			Total Industrial		%
			Total Municipal		%
			Grand Total		100 %
			Diversity		
			Total Load Demand After Diversity		

Appendix 9 – Load Estimation Sheet for Central Air Conditioning Load

(to be completed by Electricity Supply Applicant)

Contact Person: Mr/Ms _____ (Responsible Engineer/s for this application)

Company Name: _____

Telephone Number: _____

(A) Usage of Floor Area	(B) Net A/C Area (m ²)	(C) Estimated Total A/C Cooling Load (kW Cooling)	(D) Propose Type of A/C System	(E) Estimated Coefficient of Performance	(F) Estimated Total A/C Electrical Load (kW Elec.)	A/C Cooling Load Check Figures	
						(G) Cooling Load per m ² (W/m ²) [=(C)x1000/(B)]	(H) Floor Area per kW (m ² /kW) [=(B)/(C)]

Special Factors Taken into Account in Estimating the A/C Cooling Load

(I) Personnel Occupancy (m ² /person)	(J) Total area of Curtain Wall & Windows (m ²)	(K) Area with (J) facing West or South-west (m ²)	(L) Area with (B) on Top Floor (m ²)	(M) Skylight (m ²)	(N) Any Other Relevant Information?

Explanatory Notes

Preamble: Please complete this table to the best of the available design information.

(A) Please report the usage of the various portions of the building if it is a multi-usage one. The suggested classification of usage is listed below:

- Shops (individual shops fronting a street, etc.)
- Shops for rendering services (barber/beauty shop, etc.)
- Shopping centres and arcades
- Offices
- Hotels/dormitories
- Restaurants
- Theatres/auditoriums/churches
- Educational institutions
- Residential premises
- Factories – fully air-conditioned
- Factories – spot air-conditioned

(C) The unit here is kW of cooling capacity in which:

1 Tonne of Refrigeration = approx. 3.5 kW

It is a different entity from the kW of electricity in column (F)

(D) Please indicate the type of central air conditioning system such as:

- Packaged air-cooled
- Indirect (through heat exchanger) sea water cooled
- Direct sea water cooled
- Direct water cooled (cooling tower)

(E) The coefficient of performance (COP) is the estimated COP for the whole A/C installation.

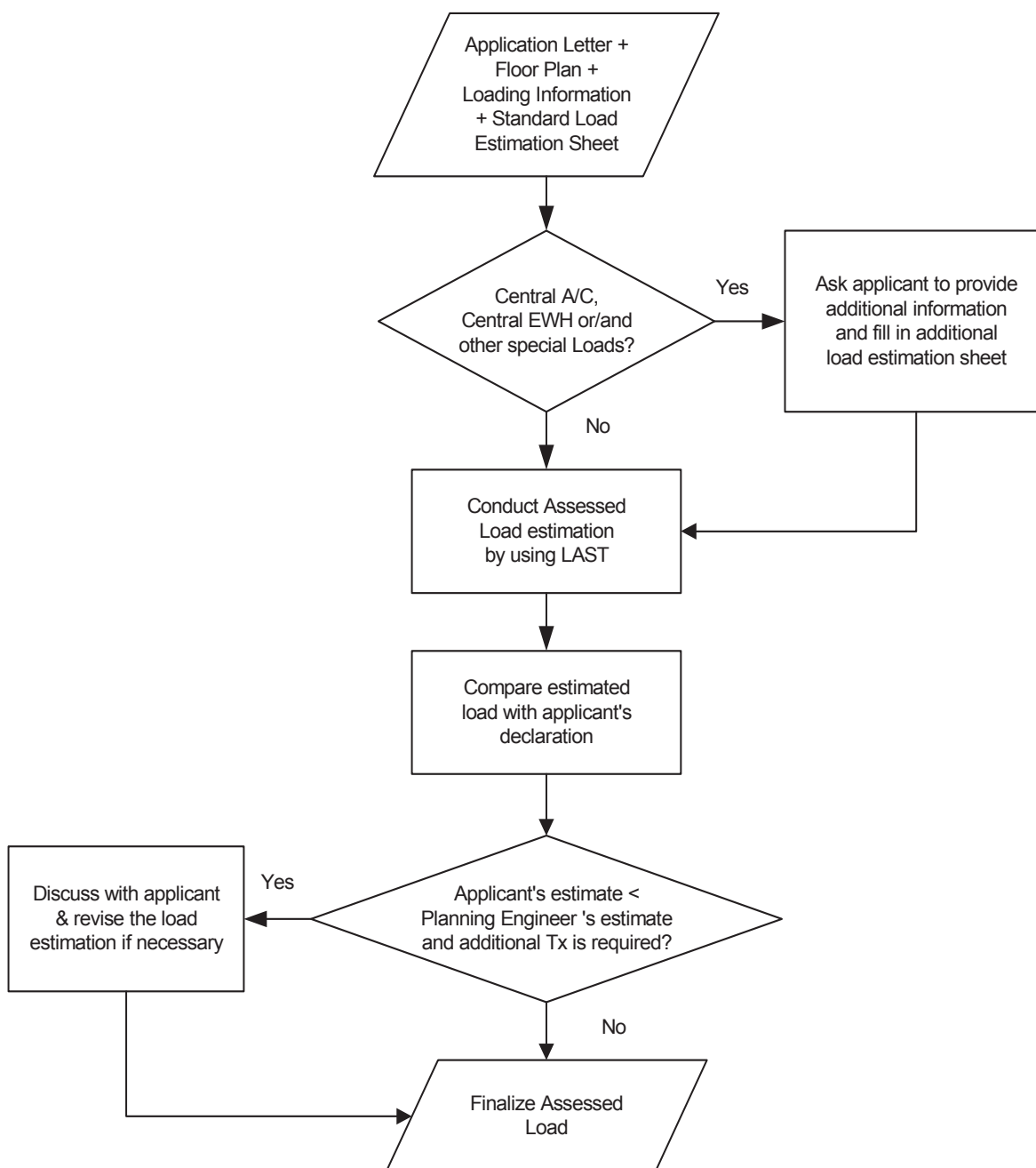
$$\text{i.e. COP} = \frac{\text{Cooling capacity of the A/C system (kW cooling) (refer to (C))}}{\text{Electrical power consumption of ALL components of the A/C installation (kW electricity) (refer to (F))}}$$

- including: - refrigeration plant such as water chillers, and chilled water pump sets
- heat rejection equipment (condensers/cooling tower/sea water pump sets)
- air-side equipment (AHU/FCU/VAU units/fans)

(F) From the equation in (E) above:

$$(F) \text{ in kW} = \frac{(C) \text{ in kW}}{\text{COP}}$$

Appendix 10 - Flow Chart for Compilation of Assessed Loads



Note: Assessed Load is used for determining both the number of transformers and the number of transformer bays.